

A. Introduction

1. **Title:** Cyber Security — Physical Security of BES Cyber Systems
2. **Number:** RRC-CIP-006-1
3. **Purpose:** To manage physical access to the Bulk Electric System (Cyber Systems (BCS) by specifying a physical security plan in support of protecting BCS compromise that could result in instability, uncontrolled separations, or Cascading within the BES.
4. **Applicability:**
 - 4.1. **Functional Entities:** For the purpose of the requirements contained herein, the following list of functional entities will be collectively referred to as “Responsible Entities.”
 - 4.1.1. **Balancing Authority**
 - 4.1.2. **Generator Operator**
 - 4.1.3. **Generator Owner**
 - 4.1.4. **Transmission Operator**
 - 4.1.5. **Transmission Owner**
 - 4.1.6. **Distribution Provider:** that owns one or more of the following Facilities, systems, and equipment for the protection or restoration of the BES:
 - 4.1.6.1. Each underfrequency load shedding (UFLS) or undervoltage loadshedding (UVLS) system that:
 - 4.1.6.1.1. is part of a Load shedding program that is subject to one or more requirements in an Railbelt Reliability Council Standard; and
 - 4.1.6.1.2. performs automatic Load shedding under a common control system owned by the Responsible Entity, without human operator initiation, that the loss, compromise, or misuse could have an adverse impact on the reliable operation of the BES.
 - 4.1.6.2. Each Remedial Action Scheme (RAS) where the RAS is subject to one or more requirements in an Railbelt Reliability Council Standard.
 - 4.1.6.3. Each Protection System (excluding UFLS and UVLS) that applies to Transmission where the Protection System is subject to one or more requirements in an Railbelt Reliability Council Standard.
 - 4.1.6.4. Each Cranking Path and group of Elements meeting the initial switching requirements from a Blackstart Resource up to and including the first interconnection point of the starting station service of the next generation unit(s) to be started.
 - 4.2. **Facilities:** For the purpose of the requirements contained herein, the following Facilities, systems, and equipment owned by each Responsible Entity in 4.1 above are those to which these requirements are applicable. For requirements in this standard where a specific type of Facilities, system, or equipment or subset of Facilities, systems, and equipment are applicable, these are specified explicitly.

4.2.1. Distribution Provider: One or more of the following Facilities, systems and equipment owned by the Distribution Provider for the protection or restoration of the BES:

4.2.1.1. Each UFLS or UVLS System that:

4.2.1.1.1. is part of a Load shedding program that is subject to one or more requirements in an Railbelt Reliability Council Standard; and

4.2.1.1.2. performs automatic Load shedding under a common control system owned by the Responsible Entity, without human operator initiation, that the loss, compromise, or misuse of could have adverse impact on the reliable operation of the BES.

4.2.1.2. Each RAS where the RAS is subject to one or more requirements in an Railbelt Reliability Council Standard.

4.2.1.3. Each Protection System (excluding UFLS and UVLS) that applies to Transmission where the Protection System is subject to one or more requirements in an Railbelt Reliability Council Standard.

4.2.1.4. Each Cranking Path and group of Elements meeting the initial switching requirements from a Blackstart Resource up to and including the first interconnection point of the starting station service of the next generation unit(s) to be started.

4.2.2. Responsible Entities listed in 4.1 other than Distribution Providers:

All BES Facilities.

4.2.3. Exemptions: The following are exempt from Standard **RRC-CIP-006-1**:

4.2.3.1. Cyber Systems at Facilities regulated by the Canadian Nuclear Safety Commission.

4.2.3.2. Cyber Assets associated with communication networks and data communication links between discrete Electronic Security Perimeters (ESP).

4.2.3.3. Cyber Systems, associated with communication networks and data communication links, between Cyber Systems, providing confidentiality and integrity of an ESP that extends to one or more geographic locations.

4.2.3.4. The systems, structures, and components that are regulated by the Nuclear Regulatory Commission under a cyber security plan pursuant to 10 C.F.R. Section 73.54.

4.2.3.5. For Distribution Providers, the systems and equipment that are not included in section 4.2.1 above.

4.2.3.6. Responsible Entities that identify that they have no BES Cyber Systems categorized as high impact or medium impact according to the CIP-002 identification and categorization processes.

4.3. “Applicable Systems”: Each table has an “Applicable Systems” column to define the scope of systems to which a specific Requirement Part applies.

5. Effective Dates: Shall become effective on the first day of the first calendar quarter that is thirty-six (36) months following adoption by the Approving Authority.

B. Requirements and Measures

- R1.** Each Responsible Entity shall implement one or more documented physical security plan(s) that collectively include all of the applicable requirement parts in RRC-CIP-006-1 *Table R1 – Physical Security Plan*. *[Violation Risk Factor: Medium]*
- M1.** Evidence must include each of the documented physical security plans that collectively include all of the applicable requirement parts in RRC-CIP-006-1 *Table R1 – Physical Security Plan* and additional evidence to demonstrate implementation of the plan or plans as described in the Measures column of the table.

RRC-CIP-006-1 Table R1 – Physical Security Plan			
Part	Applicable Systems	Requirements	Measures
1.1	Medium Impact BCS without External Routable Connectivity (ERC) Physical Access Control Systems (PACS) associated with: • High Impact BCS, or • Medium Impact BCS with ERC SCI supporting an Applicable System in this Part	Define operational or procedural controls to restrict physical access.	An example of evidence may include, but is not limited to, documentation that operational or procedural controls exist.

RRC-CIP-006-1 Table R1 – Physical Security Plan			
Part	Applicable Systems	Requirements	Measures
1.2	Medium Impact BCS with ERC and their associated: 1. EACMS; and 2. Protected Cyber Assets (PCA) SCI supporting an Applicable System in this Part	Utilize at least one physical access control to allow unescorted physical access into each applicable Physical Security Perimeter (PSP) to only those individuals who have authorized unescorted physical access.	An example of evidence may include, but is not limited to, language in the physical security plan that describes each PSP and how unescorted physical access is controlled by one or more different methods and proof that unescorted physical access is restricted to only authorized individuals, such as a list of authorized individuals accompanied by access logs.
1.3	High Impact BCS and their associated: 1. EACMS; and 2. PCA SCI supporting an Applicable System in this Part	Where capable, utilize two or more different physical access controls (this does not require two completely independent physical access control systems) to collectively allow unescorted physical access into PSP to only those individuals who have authorized unescorted physical access.	Examples of evidence may include, but is not limited to, language in the physical security plan that describes the PSP and how unescorted physical access is controlled by two or more different methods and proof that unescorted physical access is restricted to only authorized individuals, such as a list of authorized individuals accompanied by access logs.

RRC-CIP-006-1 Table R1 – Physical Security Plan			
Part	Applicable Systems	Requirements	Measures
1.4	High Impact BCS and their associated: 1. EACMS; and 2. PCA Medium Impact BCS with ERC and their associated: 1. EACMS; and 2. PCA SCI supporting an Applicable System in this Part	Monitor for unauthorized access through a physical access point into a PSP.	An example of evidence may include, but is not limited to, documentation of controls that monitor for unauthorized access through a physical access point into a PSP.
1.5	High Impact BCS and their associated: 1. EACMS; and 2. PCA Medium Impact BCS with ERC and their associated: 1. EACMS; and 2. PCA SCI supporting an Applicable System in this Part	Issue an alarm or alert in response to detected unauthorized access through a physical access point into a PSP to the personnel identified in the Cyber Security Incident response plan within 15 minutes of detection.	Examples of evidence may include, but is not limited to, language in the physical security plan that describes the issuance of an alarm or alert in response to unauthorized access through a physical access control into a PSP and additional evidence that the alarm or alert was issued and communicated as identified in the BES Cyber Security Incident Response Plan, such as manual or electronic alarm or alert logs, cell phone or pager logs, or other evidence that documents that the alarm or alert was generated and communicated.

RRC-CIP-006-1 Table R1 – Physical Security Plan			
Part	Applicable Systems	Requirements	Measures
1.6	Physical Access Control Systems (PACS) associated with: • High Impact BCS, or • Medium Impact BCS with ERC SCI supporting an Applicable System in this Part	Monitor each PACS for unauthorized physical access to a PACS	An example of evidence may include, but is not limited to, documentation of controls that monitor for unauthorized physical access to a PACS.
1.7	Physical Access Control Systems (PACS) associated with: • High Impact BCS, or • Medium Impact BCS with ERC SCI supporting an Applicable System in this Part	Issue an alarm or alert in response to detected unauthorized physical access to a PACS to the personnel identified in the Cyber Security Incident response plan within 15 minutes of the detection.	Examples of evidence may include, but is not limited to, language in the physical security plan that describes the issuance of an alarm or alert in response to unauthorized physical access to PACS and additional evidence that the alarm or alerts was issued and communicated as identified in the BES Cyber Security Incident Response Plan, such as alarm or alert logs, cell phone or pager logs, or other evidence that the alarm or alert was generated and communicated.

RRC-CIP-006-1 Table R1 – Physical Security Plan			
Part	Applicable Systems	Requirements	Measures
1.8	High Impact BCS and their associated: 1. EACMS; and 2. PCA 3. Medium impact BCS with ERC and their associated: 1. EACMS; and 2. PCA SCI supporting an Applicable System in this Part	Log (through automated means or by personnel who control entry) entry of each individual with authorized unescorted physical access into each PSP, with information to identify the individual and date and time of entry.	Examples of evidence may include, but is not limited to, language in the physical security plan that describes logging and recording of physical entry into each PSP and additional evidence to demonstrate that this logging has been implemented, such as logs of physical access into PSP that show the individual and the date and time of entry into PSP.
1.9	High Impact BCS and their associated: 1. EACMS; and 2. PCA SCI supporting an Applicable System in this Part Medium impact BCS with ERC and their associated: 1. EACMS; and 2. PCA SCI supporting an Applicable System in this Part	Retain physical access logs of entry of individuals with authorized unescorted physical access into each PSP for at least 90 calendar days.	An example of evidence may include, but is not limited to, dated documentation such as logs of physical access into PSP that show the date and time of entry into PSP.

- R2.** Each Responsible Entity shall implement one or more documented visitor control program(s) that include each of the applicable requirement parts in RRC-CIP-006-1 *Table R2 – Visitor Control Program*. [Violation Risk Factor: Medium]
- M2.** Evidence must include one or more documented visitor control programs that collectively include each of the applicable requirement parts in RRC-CIP-006-1 *Table R2 – Visitor Control Program* and additional evidence to demonstrate implementation as described in the Measures column of the table.

RRC-CIP-006-1 Table R2 – Visitor Control Program			
Part	Applicable Systems	Requirements	Measures
2.1	High Impact BCS and their associated: 1. EACMS; and 2. PCA Medium Impact BCS with ERC and their associated: 1. EACMS; and 2. PCA SCI supporting an Applicable System in this Part	Require continuous escorted access of visitors (individuals who are provided access but are not authorized for unescorted physical access) within each PSP.	An example of evidence may include, but is not limited to, language in a visitor control program that requires continuous escorted access of visitors within each PSP and additional evidence to demonstrate that the process was implemented, such as visitor logs.

RRC-CIP-006-1 Table R2 – Visitor Control Program			
Part	Applicable Systems	Requirements	Measures
2.2	High Impact BCS and their associated: 1. EACMS; and 2. PCA Medium Impact BCS with ERC and their associated: 1. EACMS; and 2. PCA SCI supporting an Applicable System in this Part	Require manual or automated logging of visitor entry into and exit from each PSP that includes date and time of the initial entry and last exit, the visitor's name, and the name of an individual point of contact responsible for the visitor.	An example of evidence may include, but is not limited to, language in a visitor control program that requires escorted access of visitors within Physical Security Perimeters and additional evidence to demonstrate that the process was implemented, such as dated visitor logs that include the required information.
2.3	High Impact BCS and their associated: 1. EACMS; and 2. PCA Medium Impact BCS with ERC and their associated: 1. EACMS; and 2. PCA SCI supporting an Applicable System in this Part	Retain visitor logs for at least 90 calendar days.	An example of evidence may include, but is not limited to, documentation showing logs have been retained for at least ninety calendar days.

- R3.** Each Responsible Entity shall implement one or more documented Physical Access Control System maintenance and testing program(s) that collectively include each of the applicable requirement parts in RRC-CIP-006-1 *Table R3 – Physical Access Control System Maintenance and Testing Program*. [Violation Risk Factor: Medium].
- M3.** Evidence must include each of the documented Physical Access Control System maintenance and testing programs that collectively include each of the applicable requirement parts in RRC-CIP-006-1 *Table R3 – Physical Access Control System Maintenance and Testing Program* and additional evidence to demonstrate implementation as described in the Measures column of the table.

RRC-CIP-006-1 Table R3 – Physical Access Control System Maintenance and Testing Program			
Part	Applicable Systems	Requirements	Measures
3.1	Physical Access Control Systems (PACS) associated with: • High Impact BCS, or • Medium Impact BCS with ERC Locally mounted hardware or devices at the PSP associated with: • High Impact BCS, or • Medium Impact BCS with ERC	Maintenance and testing of each PACS and locally mounted hardware or devices at the PSP at least once every 24 calendar months to ensure they function properly.	Examples of evidence may include, but is not limited to, a maintenance and testing program that provides for testing each PACS and locally mounted hardware or devices associated with each applicable PSP at least once every 24 calendar months and additional evidence to demonstrate that this testing was done, such as dated maintenance records, or other documentation showing testing and maintenance has been performed on each applicable device or system at least once every 24 calendar months.

C. Compliance

1 Compliance Monitoring Process

- 1.1. Compliance Enforcement Authority:** The RRC is the entity with the authority to monitor and enforce compliance with mandatory and enforceable Railbelt Reliability Council Standards.
- 1.2. Evidence Retention:** The following evidence retention periods identify the period of time an entity is required to retain specific evidence to demonstrate compliance. For instances where the evidence retention period specified below is shorter than the time since the last audit, the RRC may ask an entity to provide other evidence to show that it was compliant for the full time period since the last audit.

The Responsible Entity shall keep data or evidence to show compliance as identified below unless directed by the RRC to retain specific evidence for a longer period of time as part of an investigation:

- Each Responsible Entity shall retain evidence of each requirement in this standard for three calendar years.
 - If a Responsible Entity is found non-compliant, it shall keep information related to the non-compliance until mitigation is complete and approved or the time specified above, whichever is longer.
 - The RRC shall keep the last audit records and all requested and submitted subsequent audit records.
- 1.3. Compliance Monitoring and Enforcement Program:** Compliance Monitoring and Enforcement Program refers to the identification of the processes that will be used to evaluate data or information for the purpose of assessing performance or outcomes with the associated Railbelt Reliability Council Standards.

Table of Compliance Elements

R#	Violation Risk Factor (VRF)	Violation Severity Levels (RRC-CIP-006-1)			
		Lower VSL	Moderate VSL	High VSL	Severe VSL
R1	Medium	N/A	N/A	N/A	The Responsible Entity did not document or implement physical security plans. (Requirement R1) OR The Responsible Entity did not document or implement operational or procedural controls to restrict physical access. (R 1.1) OR The Responsible Entity has documented and implemented physical access controls, but at least one control does not exist to restrict access to Applicable Systems. (R 1.2) OR The Responsible Entity has documented and implemented physical access controls, but at least two different controls do not exist

R#	Violation Risk Factor (VRF)	Violation Severity Levels (RRC-CIP-006-1)			
		Lower VSL	Moderate VSL	High VSL	Severe VSL
					to restrict access to Applicable Systems. (R 1.3) OR The Responsible Entity does not have a process to monitor for unauthorized access through a physical access point into a PSP. (R 1.4) OR The Responsible Entity does not have a process to alert for detected unauthorized access through a physical access point into a PSP or to communicate such alerts within 15 minutes to identified personnel. (R 1.5) OR The Responsible Entity does not have a process to monitor each PACS for unauthorized physical access to a PACS. (R 1.6) OR

R#	Violation Risk Factor (VRF)	Violation Severity Levels (RRC-CIP-006-1)			
		Lower VSL	Moderate VSL	High VSL	Severe VSL
					The Responsible Entity does not have a process to alert for unauthorized physical access to PACS or to communicate such alerts within 15 minutes to identified personnel. (R 1.7) OR The Responsible Entity does not have a process to log authorized physical entry into each PSP with sufficient information to identify the individual and date and time of entry. (R 1.8) OR The Responsible Entity does not have a process to retain physical access logs for 90 calendar days. (R 1.9)
R2	Medium	N/A	N/A	N/A	The Responsible Entity has failed to include or implement a visitor control program that requires continuous escorted access of visitors within any Physical Security Perimeter. (R 2.1) OR

R#	Violation Risk Factor (VRF)	Violation Severity Levels (RRC-CIP-006-1)			
		Lower VSL	Moderate VSL	High VSL	Severe VSL
					The Responsible Entity has failed to include or implement a visitor control program that requires logging of the initial entry and last exit dates and times of the visitor, the visitor's name, and the point of contact. (R 2.2) OR The Responsible Entity failed to include or implement a visitor control program to retain visitor logs for at least 90 days. (R 2.3)
R3	Medium	The Responsible Entity has documented and implemented a maintenance and testing program for Physical Access Control Systems and locally mounted hardware or devices at the Physical Security Perimeter, but did not complete required testing greater than 24 calendar months but less than or equal to 26 calendar months. (R 3.1)	The Responsible Entity has documented and implemented a maintenance and testing program for Physical Access Control Systems and locally mounted hardware or devices at the PSP, but did not required testing greater than 26 calendar months but less than or equal to 28 calendar months. (R 3.1)	The Responsible Entity has documented and implemented a maintenance and testing program for PACS and locally mounted hardware or devices at the Physical Security Perimeter, but did not complete required testing greater than 28 calendar months but less than or equal to 30 calendar months. (R 3.1)	The Responsible Entity did not document or implement a maintenance and testing program for PACS and locally mounted hardware or devices at the PSP. (R 3.1) OR The Responsible Entity has documented and implemented a maintenance and testing program for PACS and locally mounted

R#	Violation Risk Factor (VRF)	Violation Severity Levels (RRC-CIP-006-1)			
		Lower VSL	Moderate VSL	High VSL	Severe VSL
					hardware or devices at the PSP, but did not complete required testing greater than 30 calendar months. (R 3.1)

D. Regional Variances

None.

E. Interpretations

None.

F. Associated Documents

None.

Version History

Version	Date	Action	Change Tracking

Glossary

Balancing Authority	The responsible entity that integrates resource plans ahead of time, maintains Demand and resource balance within a Balancing Authority Area, and supports Interconnection frequency in real time.
BES Cyber Asset (BCA)	A Cyber Asset that if rendered unavailable, degraded, or misused would, within 15 minutes of its required operation, misoperation, or non-operation, adversely impact one or more Facilities, systems, or equipment, which, if destroyed, degraded, or otherwise rendered unavailable when needed, would affect the reliable operation of the Bulk Electric System. Redundancy of affected Facilities, systems, and equipment shall not be considered when determining adverse impact. Each BES Cyber Asset is included in one or more BES Cyber Systems.
BES Cyber System (BCS)	One or more BES Cyber Assets logically grouped by a responsible entity to perform one or more reliability tasks for a functional entity.
BES Cyber System Information	Information about the BES Cyber System that could be used to gain unauthorized access or pose a security threat to the BES Cyber System. BES Cyber System Information does not include individual pieces of information that by themselves do not pose a threat or could not be used to allow unauthorized access to BES Cyber Systems, such as, but not limited to, device names, individual IP addresses without context, ESP names, or policy statements. Examples of BES Cyber System Information may include, but are not limited to, security procedures or security information about BES Cyber Systems, Physical Access Control Systems, and Electronic Access Control or Monitoring Systems that is not publicly available and could be used to allow unauthorized access or unauthorized distribution; collections of network addresses; and network topology of the BES Cyber System.
Blackstart Resource	A generating unit(s) and its associated set of equipment which has the ability to be started without support from the System or is designed to remain energized without connection to the remainder of the System, with the ability to energize a bus, meeting the Transmission Operator's restoration plan needs for Real and Reactive Power capability, frequency and voltage control, and that has been included in the Transmission Operator's restoration plan.
Bulk Electric System (BES)	Unless modified by the lists shown below, all Transmission Elements operated at 100 kV or higher and Real Power and Reactive Power resources connected at 100 kV or higher. This does not include facilities used in the local distribution of electric energy. Inclusions: • I1 - Transformers with the primary terminal and at least one secondary terminal operated at 100 kV or higher unless excluded by application of Exclusion E1 or E3. • I2 – Generating resource(s) including the generator terminals through the

high-side of the

step-up transformer(s) connected at a voltage of 100 kV or above with:

- a) Gross individual nameplate rating greater than 20 MVA. Or,
- b) Gross plant/facility aggregate nameplate rating greater than 75 MVA.

- I3 - Blackstart Resources identified in the Transmission Operator's restoration plan.

- I4 - Dispersed power producing resources that aggregate to a total capacity greater than 75 MVA (gross nameplate rating), and that are connected through a system designed primarily for delivering such capacity to a common point of connection at a voltage of 100 kV or above. Thus, the facilities designated as BES are:

- a) The individual resources, and

The system designed primarily for delivering capacity from the point where those resources aggregate to greater than 75 MVA to a common point of connection at a voltage of 100 kV or above.

- I5 - Static or dynamic devices (excluding generators) dedicated to supplying or absorbing

Reactive Power that are connected at 100 kV or higher, or through a dedicated transformer with a high-side voltage of 100 kV or higher, or through a transformer that is designated in Inclusion I1 unless excluded by application of Exclusion E4

Exclusions:

- E1 - Radial systems: A group of contiguous transmission Elements that emanates from a single point of connection of 100 kV or higher and:

- a) Only serves Load. Or,
- b) Only includes generation resources, not identified in Inclusions I2, I3, or I4, with an aggregate capacity less than or equal to 75 MVA (gross nameplate rating). Or,
- c) Where the radial system serves Load and includes generation resources, not identified in Inclusions I2, I3 or I4, with an aggregate capacity of non-retail generation less than or equal to 75 MVA (gross nameplate rating).

Note 1 – A normally open switching device between radial systems, as depicted on prints or one-line diagrams for example, does not affect this exclusion.

Note 2 – The presence of a contiguous loop, operated at a voltage level of 50 kV or less, between configurations being considered as radial systems, does not affect this exclusion.

- E2 - A generating unit or multiple generating units on the customer's side of the retail meter that serve all or part of the retail Load with electric energy if: (i) the net capacity provided to the BES does not exceed 75 MVA, and (ii) standby, back-up, and maintenance power services are provided to the generating unit or multiple generating units or to the retail Load by a Balancing Authority, or provided pursuant to a binding obligation with a Generator Owner or Generator Operator, or under terms approved by the applicable regulatory authority.

	• E3 - Local networks (LN): A group of contiguous transmission Elements operated at less than 300 kV that distribute power to Load rather than transfer bulk power across the interconnected system. LN's emanate from multiple points of connection at 100 kV or higher to improve the level of service to retail customers and not to accommodate bulk power transfer across the interconnected system. The LN is characterized by all of the following: a) Limits on connected generation: The LN and its underlying Elements do not include generation resources identified in Inclusions I2, I3, or I4 and do not have an aggregate capacity of non-retail generation greater than 75 MVA (gross nameplate rating); b) Real Power flows only into the LN and the LN does not transfer energy originating outside the LN for delivery through the LN; and c) Not part of a Flowgate or transfer path: The LN does not contain any part of a permanent Flowgate in the Eastern Interconnection, a major transfer path within the Western Interconnection, or a comparable monitored Facility in the ERCOT or Quebec Interconnections, and is not a monitored Facility included in an Interconnection Reliability Operating Limit (IROL). • E4 – Reactive Power devices installed for the sole benefit of a retail customer(s). Note - Elements may be included or excluded on a case-by-case basis through the Rules of Procedure exception process.
CIP Exceptional Circumstance	A situation that involves or threatens to involve one or more of the following, or similar, conditions that impact safety or BES reliability: a risk of injury or death; a natural disaster; civil unrest; an imminent or existing hardware, software, or equipment failure; a Cyber Security Incident requiring emergency assistance; a response by emergency services; the enactment of a mutual assistance agreement; or an impediment of large scale workforce availability.
CIP Senior Manager	
Control Center	One or more facilities hosting operating personnel that monitor and control the Bulk Electric System (BES) in real-time to perform the reliability tasks, including their associated data centers, of: 1) a Reliability Coordinator, 2) a Balancing Authority, 3) a Transmission Operator for transmission Facilities at two or more locations, or 4) a Generator Operator for generation Facilities at two or more locations.
Cranking Path	A portion of the electric system that can be isolated and then energized to deliver electric power from a generation source to enable the startup of one or more other generating units.
Cyber Assets	Programmable electronic devices, including the hardware, software, and data in those devices.

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Distribution Provider	Provides and operates the “wires” between the transmission system and the end-use customer. For those end-use customers who are served at transmission voltages, the Transmission Owner also serves as the Distribution Provider. Thus, the Distribution Provider is not defined by a specific voltage, but rather as performing the distribution function at any voltage
Electronic Access Control or Monitoring Systems	Cyber Assets that perform electronic access control or electronic access monitoring of the Electronic Security Perimeter(s) or BES Cyber Systems. This includes Intermediate Systems.
Electronic Access Point	A Cyber Asset interface on an Electronic Security Perimeter that allows routable communication between Cyber Assets outside an Electronic Security Perimeter and Cyber Assets inside an Electronic Security Perimeter.
Electronic Security Perimeter	The logical border surrounding a network to which BES Cyber Systems are connected using a routable protocol.
Element	Any electrical device with terminals that may be connected to other electrical devices such as a generator, transformer, circuit breaker, bus section, or transmission line. An Element may be comprised of one or more components.
External Routable Connectivity	Programmable electronic devices, excluding Shared Cyber Infrastructure, including the hardware, software, and data in those devices. Application containers are considered software of Virtual Cyber Assets (VCAs) or Cyber Assets. VCAs are not considered software or data of Cyber Assets.
Interactive Remote Access	User-initiated access by a person employing a remote access client or other remote access technology using a routable protocol. Remote access originates from a Cyber Asset that is not an Intermediate System and not located within any of the Responsible Entity’s Electronic Security Perimeter(s) or at a defined Electronic Access Point (EAP). Remote access may be initiated from: 1) Cyber Assets used or owned by the Responsible Entity, 2) Cyber Assets used or owned by employees, and 3) Cyber Assets
Physical Access Control Systems	Cyber Assets that control, alert, or log access to the Physical Security Perimeter(s), exclusive of locally mounted hardware or devices at the Physical Security Perimeter such as motion sensors, electronic lock control mechanisms, and badge readers.
Physical Security Perimeter	The physical border surrounding locations in which BES Cyber Assets, BES Cyber Systems, or Electronic Access Control or Monitoring Systems reside, and for which access is controlled.
Protected Cyber Assets	One or more Cyber Assets connected using a routable protocol within or on an Electronic Security Perimeter that is not part of the highest impact BES Cyber System within the same Electronic Security Perimeter. The impact

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	rating of Protected Cyber Assets is equal to the highest rated BES Cyber System in the same ESP.
Shared Cyber Infrastructure	One or more programmable electronic devices, including the software that shares the devices' resources, that: • Hosts one or more Virtual Cyber Assets (VCA) included in a BES Cyber Systems (BCS) or their associated Electronic Access Control or Monitoring Systems (EACMS) or Physical Access Control Systems (PACS); and hosts one or more VCAs that are not included in, or associated with, BCS of the same impact categorization; or • Provides storage resources required for system functionality of one or more Cyber Assets or VCAs included in a BCS or their associated EACMS or PACS; and also for one or more Cyber Assets or VCAs that are not included in, or associated with, BCS of the same impact categorization. SCI does not include the supported VCAs or Cyber Assets with which it shares its resources.